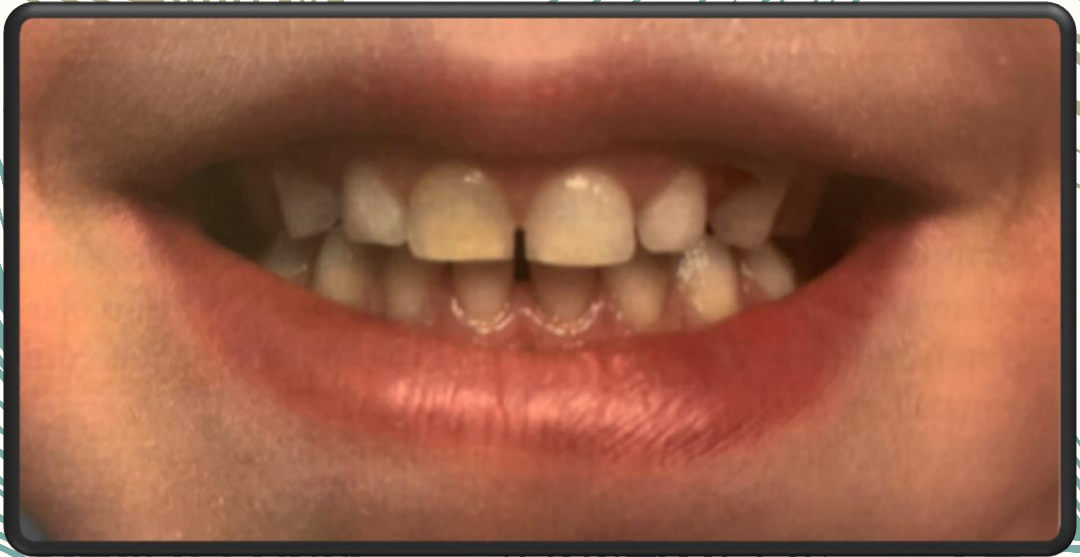

MORPHOLOGY OF DECIDUOUS ANTERIORS

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DEPT. OF ORAL BIOLOGY

DIKIOHS, DUHS

Learning Objectives:

- Identify all deciduous teeth on models
- Differentiate between primary and permanent teeth
- Explain the landmarks of all deciduous teeth with comparison
- Compare incisors and canines regarding their macroscopic structure

How is primary dentition different from permanent?

Terminologies Related to different types of dentition

Diphyodont	Having 2 sets of teeth
Deciduous	The term deciduous means literally to fall off.
Heterodont	Teeth have different morphologies (human).

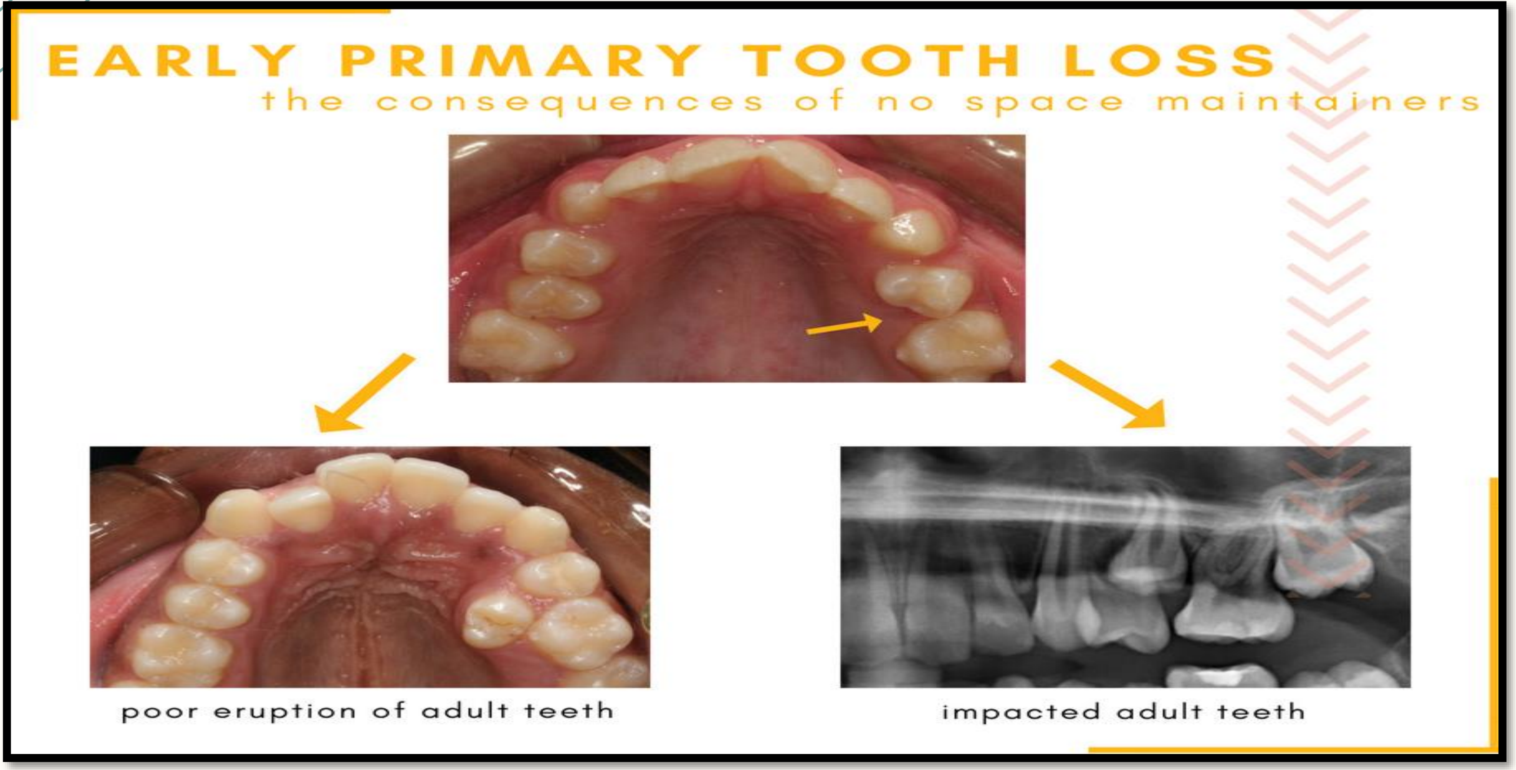
EATING AND NUTRITION:

Effective chewing & mastication → undernourishment & dietary deficiencies prevention

SELF-CONFIDENCE: boosts social interaction with confidence with teeth that are uncrowded & not even ugly

MAINTENANCE OF SPACE:

premature loss of deciduous → loss of space & development of malocclusion



WHY PRIMARY TEETH ARE IMPORTANT?

WHY PRIMARY TEETH ARE IMPORTANT?

SPEECH PRODUCTION:

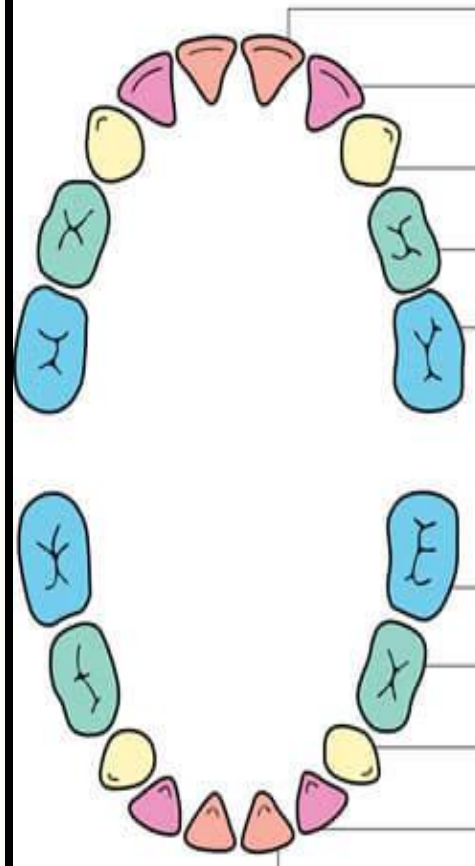
correct syllable production → tongue straying prevention during speech

PRE-CURSORS OF ORAL HEALTH MAINTAINANCE:

caries of deciduous teeth → premature loss of teeth →

- lack of spacing for permanent dentition
- defective mastication.

Fig. 3.2 Primary dentition in a child 5 years of age (Courtesy Elena Farfel, DMD, Las Vegas, NV.)



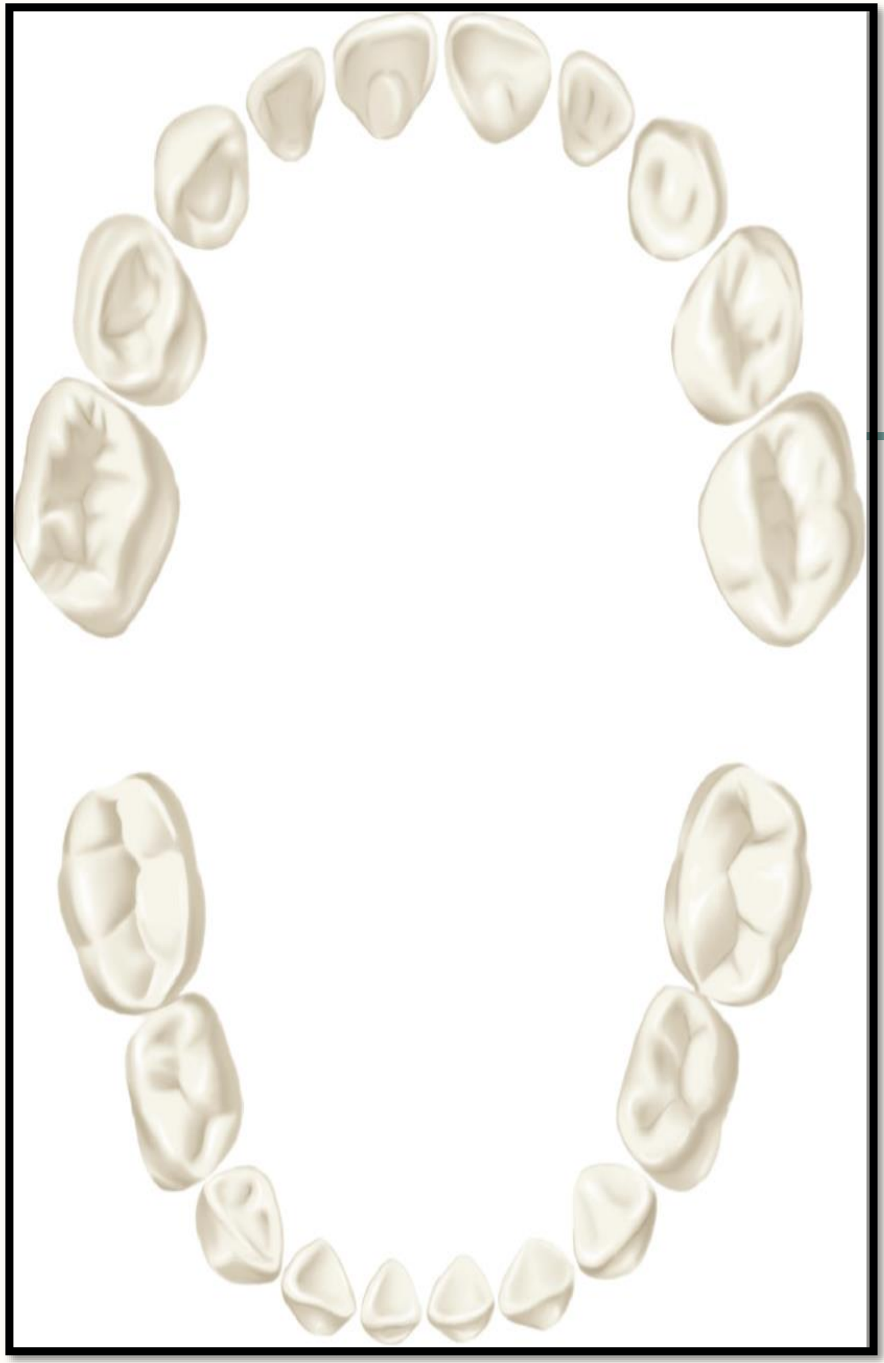
Upper Teeth	Erupt	Shed
Central incisor	8-12 mos.	6-7 yrs.
Lateral incisor	9-13 mos.	7-8 yrs.
Cuspid (canine)	16-22 mos.	10-12 yrs.
First molar	13-19 mos.	9-11 yrs.
Second molar	25-33 mos.	10-12 yrs.

Lower Teeth	Erupt	Shed
Second molar	23-31 mos.	10-12 yrs.
First molar	14-18 mos.	9-11 yrs.
Cuspid (canine)	17-23 mos.	9-12 yrs.
Lateral incisor	10-16 mos.	7-8 yrs.
Central incisor	6-10 mos.	6-7 yrs.

ORDER OF ERUPTION MATCHES ORDER OF SHEDDING:

The loss of deciduous teeth mirrors it's eruption sequence incisors -> 1st molars -> canines -> 2nd molars

mandibular pairs preceding the maxillary teeth.



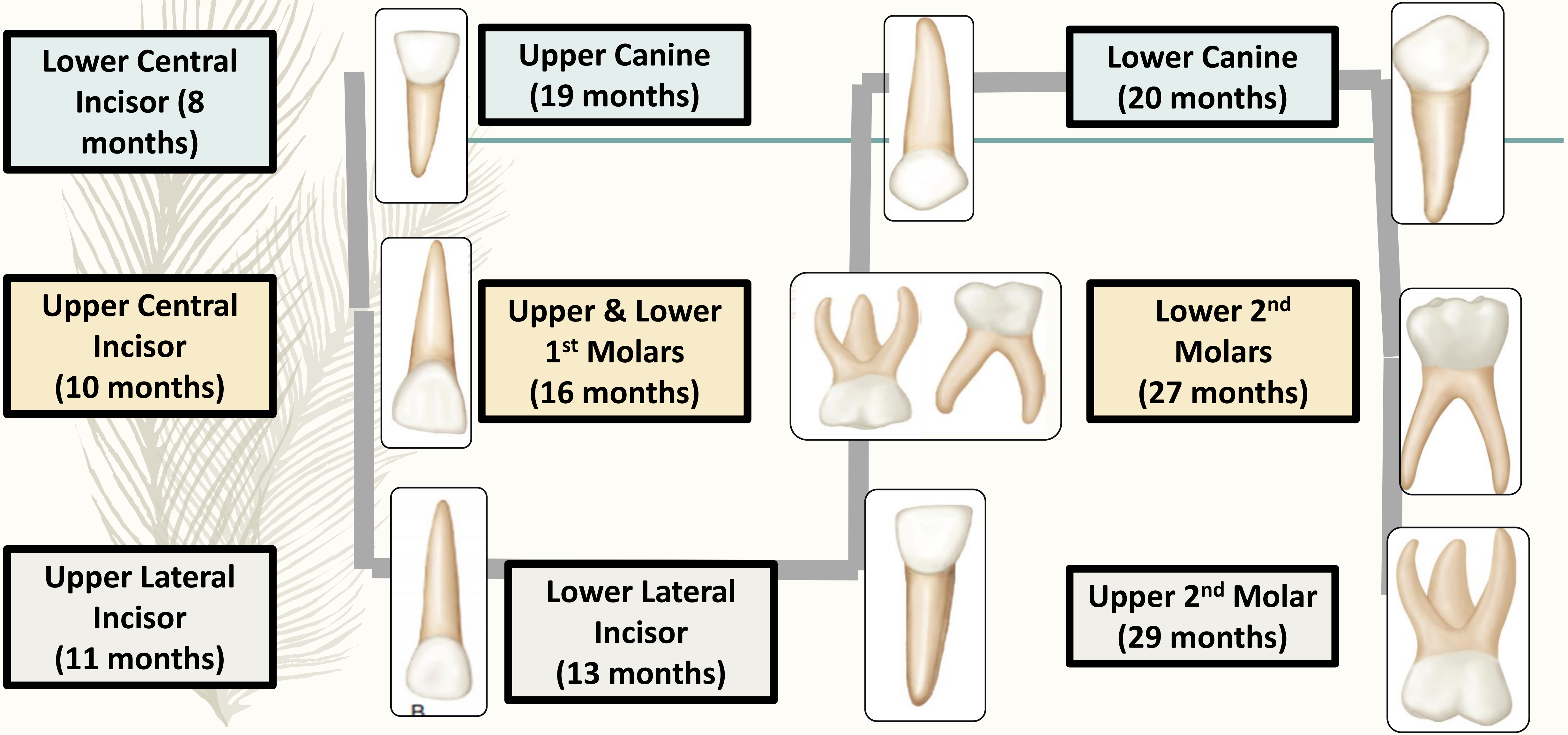
DECIDUOUS DENTITION (TEMPORARY TEETH/MILK TEETH/ BABY TEETH)

– Beginning with the median line, the teeth are named in each jaw on each side of the mouth as follows: central incisor, lateral incisor, canine, first molar, and second molar.

TOTAL # OF TEETH	20 (10 IN EACH JAW)
CLASSIFICATION	4 incisors, 2 canines, and 4 molars in each jaw.
FUNCTION	- during childhood - shed and replaced by permanent teeth. - important during growth and development for proper arrangement of the permanent teeth and proper development of the dental arches.
TIME OF COMPLETION	**2.5 YEARS OF AGE (root completion approx. 3 years)**

ORDER OF ERUPTION OF PRIMARY TEETH:

– The mandibular pairs preceding the maxillary teeth.



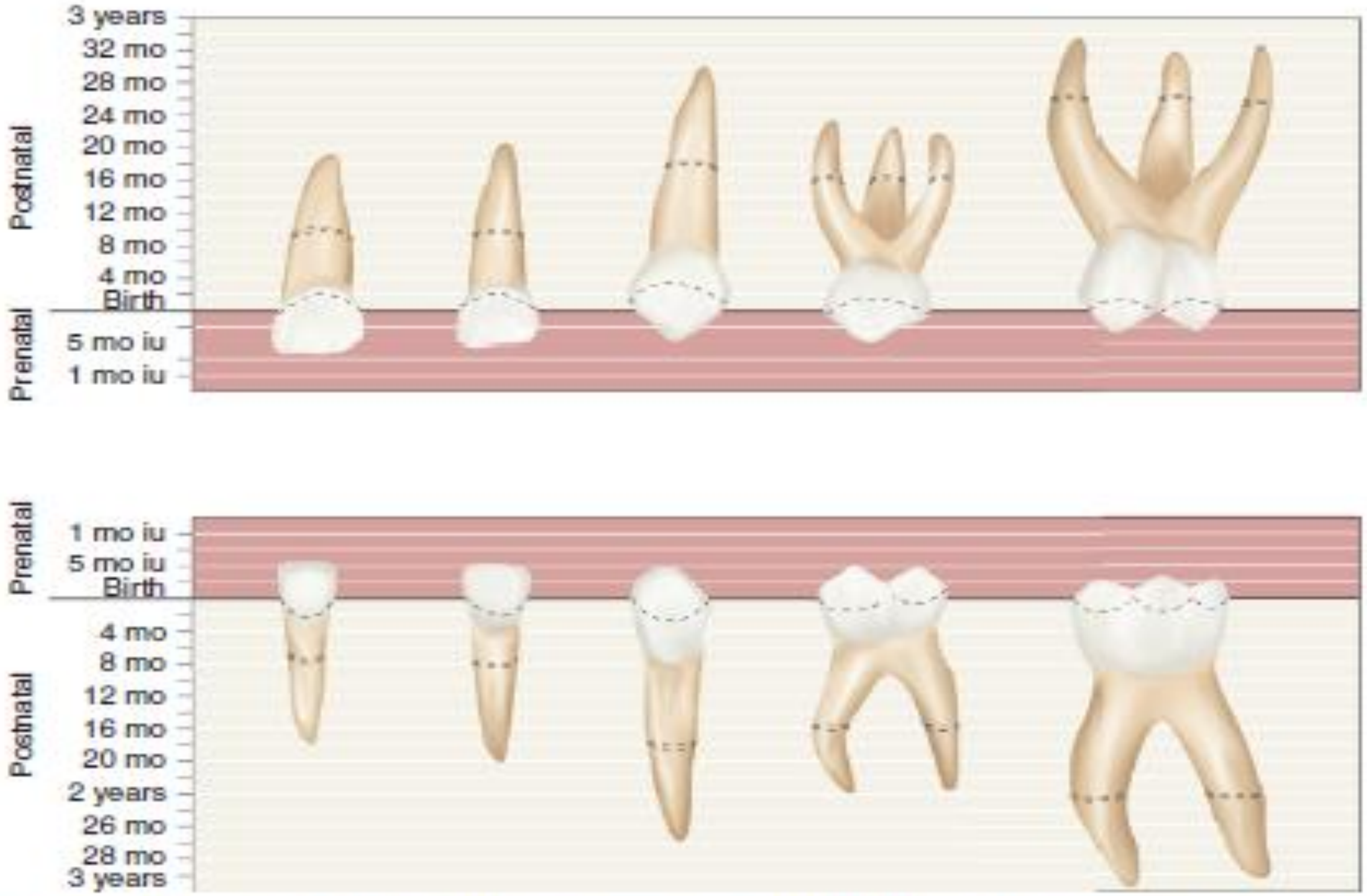
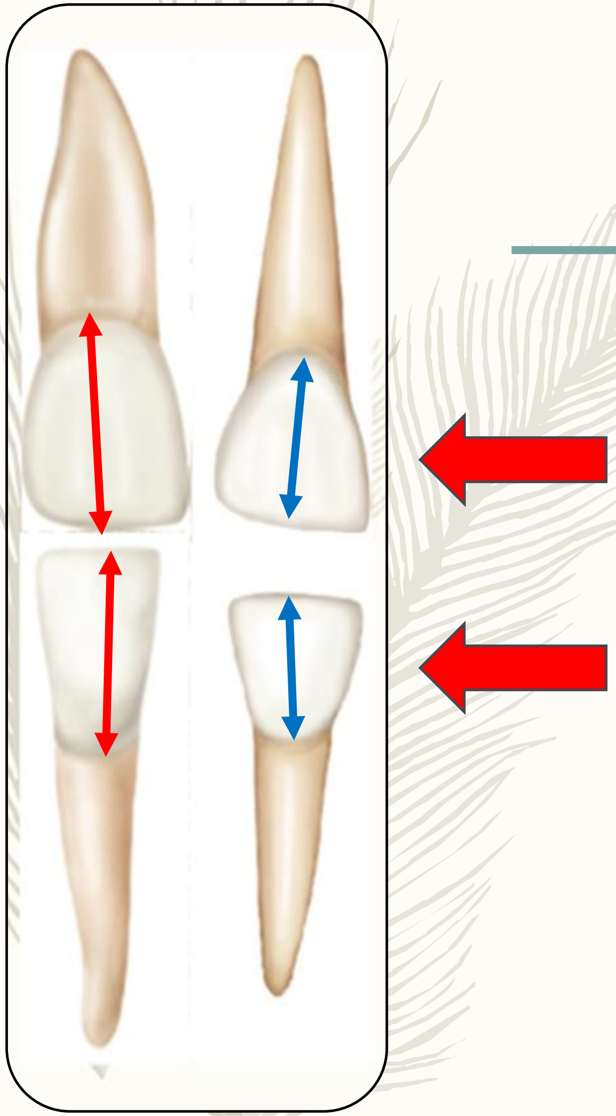


FIGURE 3-1 Diagrammatic representation of the chronology of the primary teeth. Eruption is completed at the approximate time indicated by the dotted area on the roots of the teeth. *iu*, Intrauterine.

(Modified from McBeath EC: New concept of the development and calcification of the teeth, *J Am Dent Assoc* 23:675, 1936; and Noyes EB, Shour I, Noyes HJ: *Dental histology and embryology*, ed 5, Philadelphia, 1938, Lea & Febiger.)

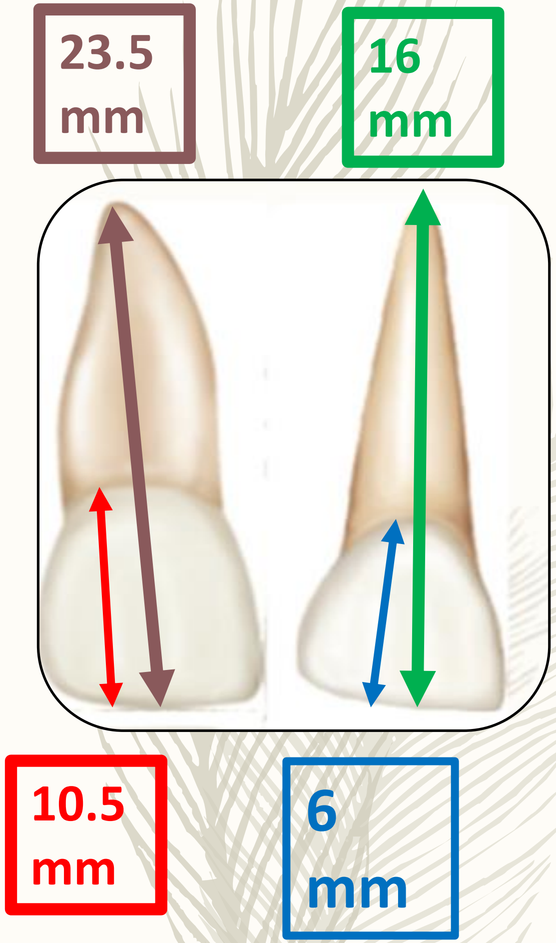
SET TRAITS (Deciduous Anteriors)



DIMENSIONS

↓ In size & crown dimensions

SET TRAITS (Deciduous Anteriors)



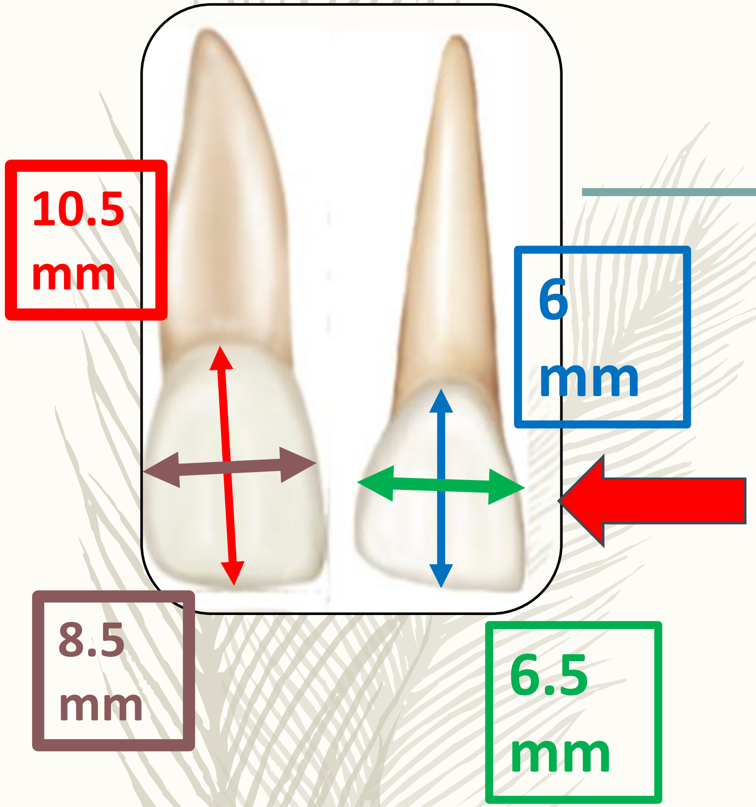
DIMENSIONS

↓ crown height/tooth length ratio

COLOR

Milk-white in color (lighter in color as compared to permanent)

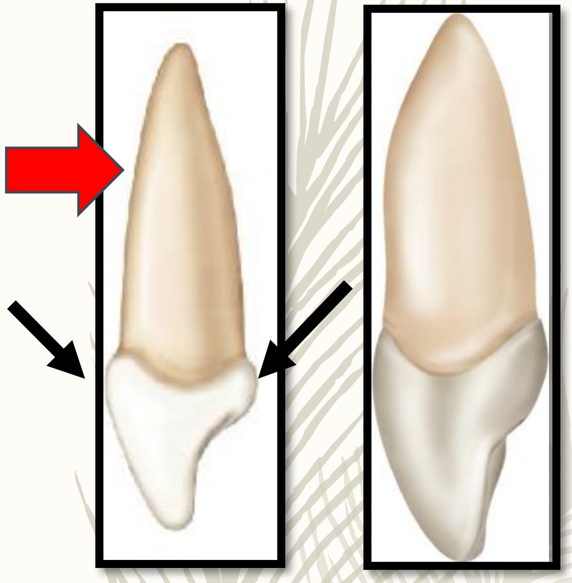
SET TRAITS (Deciduous Anteriors)

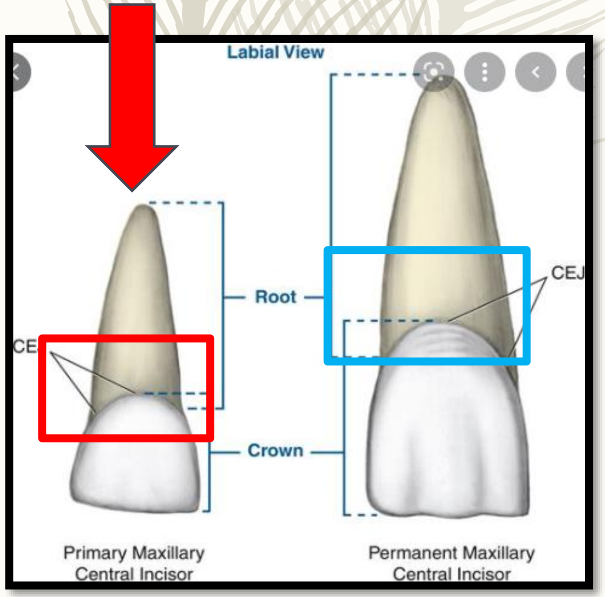


DIMENSIONS

↑ crown width/height ratio (Eg. Crowns of deciduous anterior teeth wider MD than crowns of permanent anterior teeth.)

SET TRAITS (Deciduous Anteriors)





Cervical Ridges

- More bulging B & L cervical ridges → Constricted cervix narrower at "necks"

Roots

- Roots of primary anterior teeth narrower & longer as compared to permanent.
- Narrow roots + wide crowns → arrangement at cervical 3rd of crown & root different from permanent counterparts → crown & root more slender at cervical

SET TRAITS (Deciduous Anteriors)

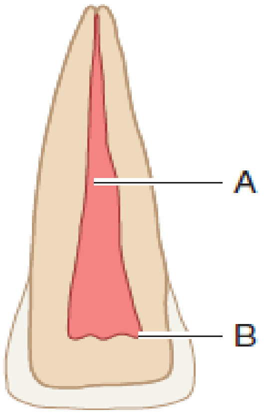


FIGURE 3-8 Permanent central incisor. **A**, Pulp canal; **B**, pulp horns. This figure represents a sectioned central incisor of a young person. Although the pulp canal is rather large, it is smaller than the pulp canal shown in Figure 3-9, and it becomes more constricted apically. Note the dentin space between the pulp horns and the incisal edge of the crown.

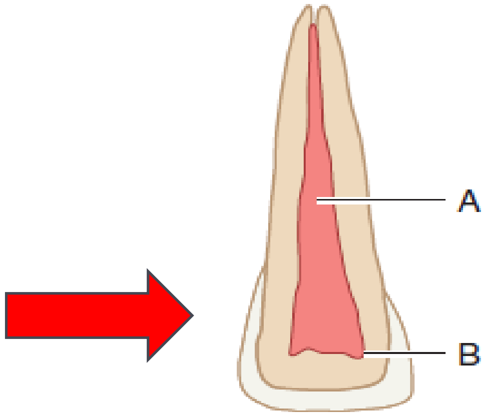


FIGURE 3-9 Primary central incisor. **A**, Pulp canal; **B**, pulp horns. This figure represents a sectioned primary central incisor. The pulp chamber with its horns and the pulp canal are broader than those found in Figure 3-8. The apical portion of the canal is much less constricted than that of the permanent tooth. Note the narrow dentin space incisally.

Cross section

- Enamel is thinner & pulp chamber is large.
- Crown widths in pulp chambers (all directions) are large as compared to root trunks & crevices (pulp canals).
- Dentin thickness limited b/w pulp chambers & enamel
- High pulp horns

PRIMATE SPACES

Interproximal spaces between primary teeth i.e between maxillary lateral incisor and canine and mandibular canine and first molar for proper alignment of teeth.

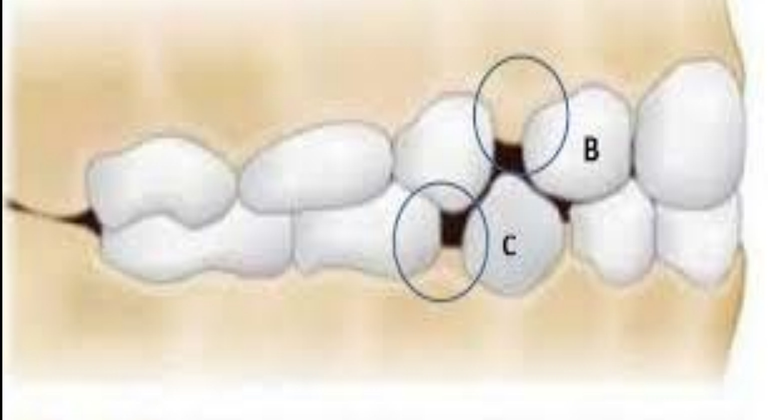


Fig. 7.16: Primate space seen in maxillary and mandibular arch

GENERALIZED/DEVELOPMENTAL OR GENERALIZED SPACING

Present in between primary teeth and play an important role in development of permanent teeth.

NEONATAL TEETH

Teeth that erupt during first month of life.

NATAL TEETH

Teeth that are present at birth above gumline.

EXFOLIATION

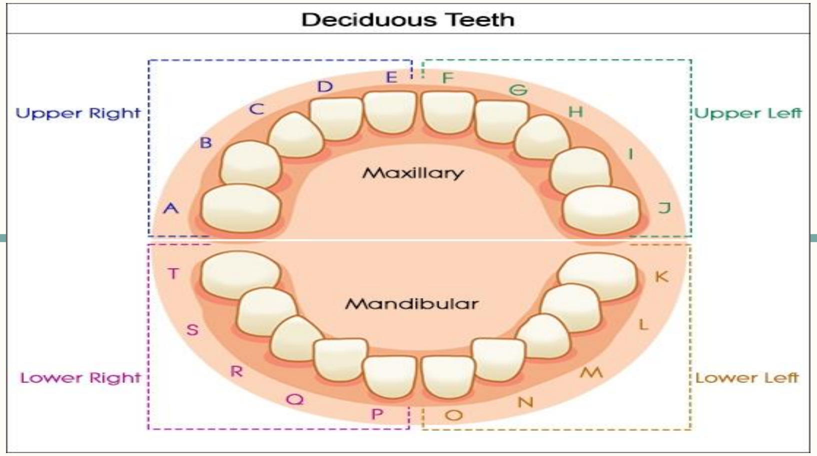
- The process of shedding of primary teeth and their replacement by permanent teeth is called **exfoliation**.

Takes place → **Between 7th & 12th years**

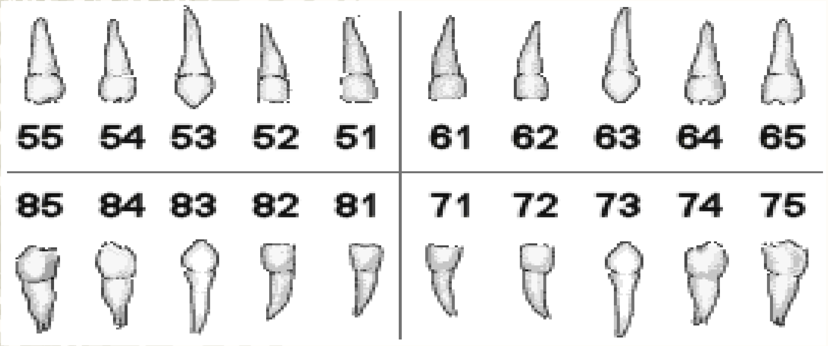
DENTAL FORMULA

$$I\frac{2}{2}C\frac{1}{1}M\frac{2}{2}=10$$

UNIVERSAL NUMBERING SYSTEM



FDI NUMBERING SYSTEM



PALMER NUMBERING SYSTEM

EDCBA	ABCDE
EDCBA	ABCDE

DECIDUOUS ANTERIORS

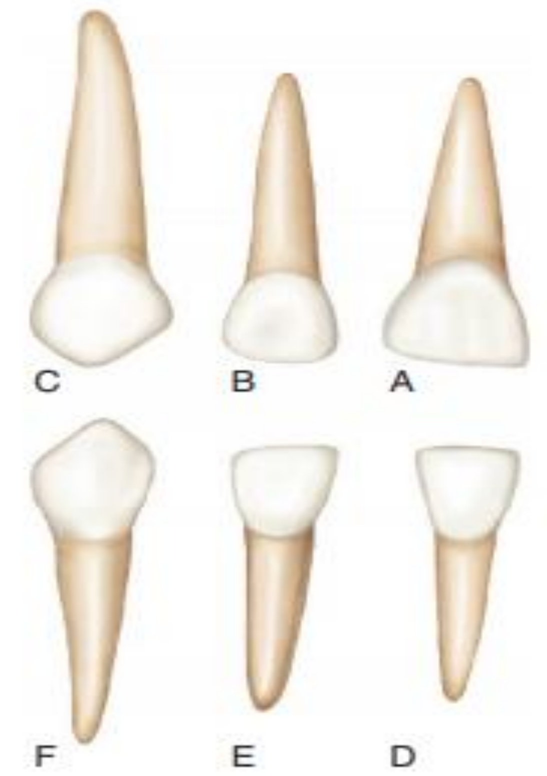
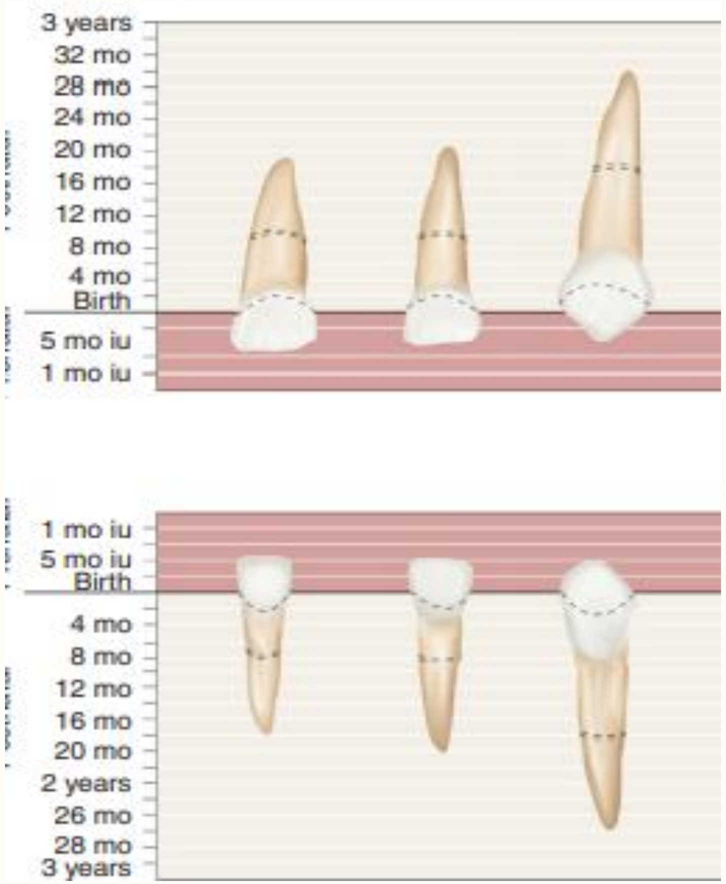
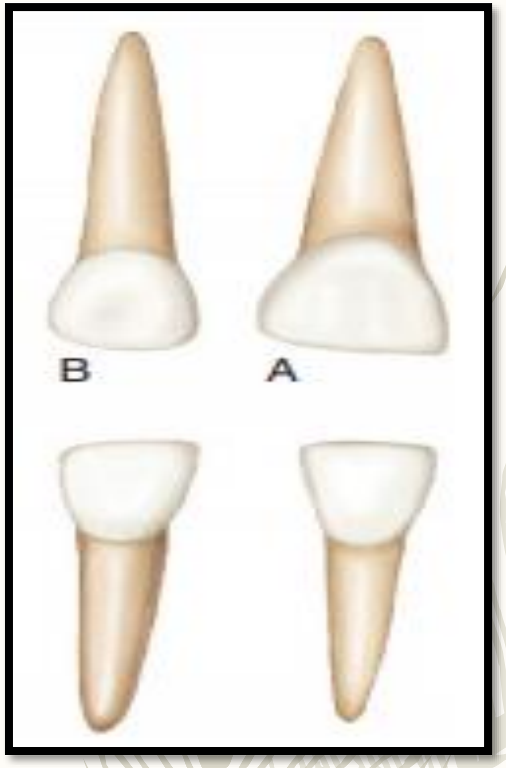
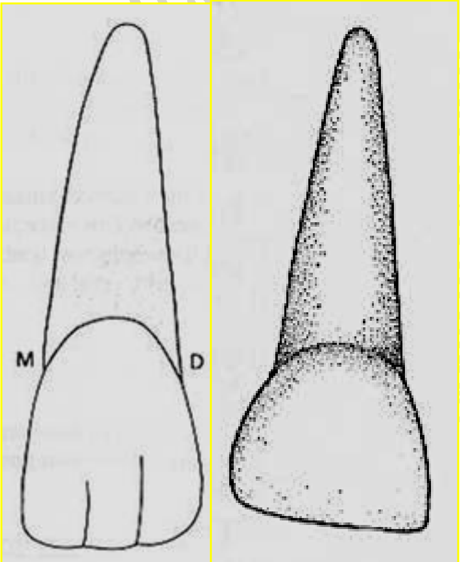


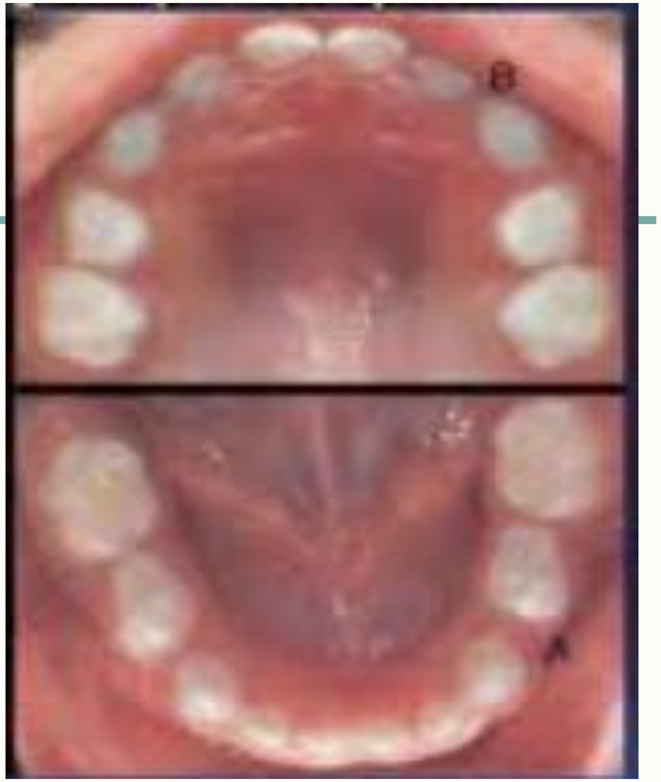
FIGURE 3-11 Primary right anterior teeth, labial aspect. A, Maxillary central incisor. B, Maxillary lateral incisor. C, Maxillary canine. D, Mandibular central incisor. E, Mandibular lateral incisor. F, Mandibular canine.



DECIDUOUS INCISORS



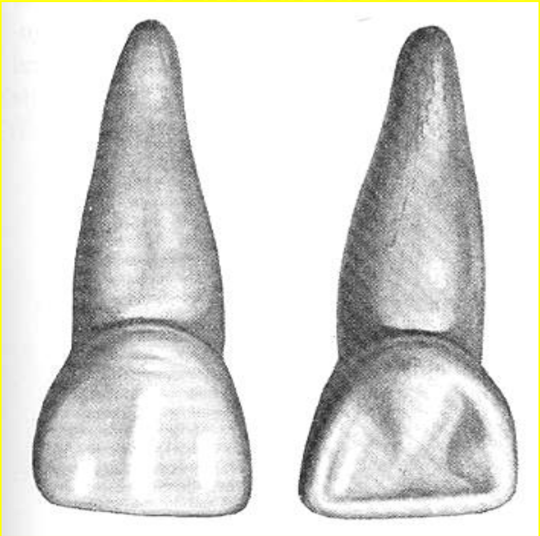


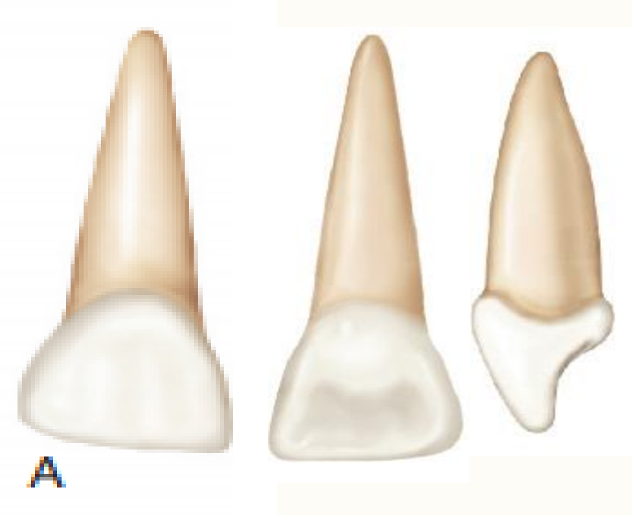


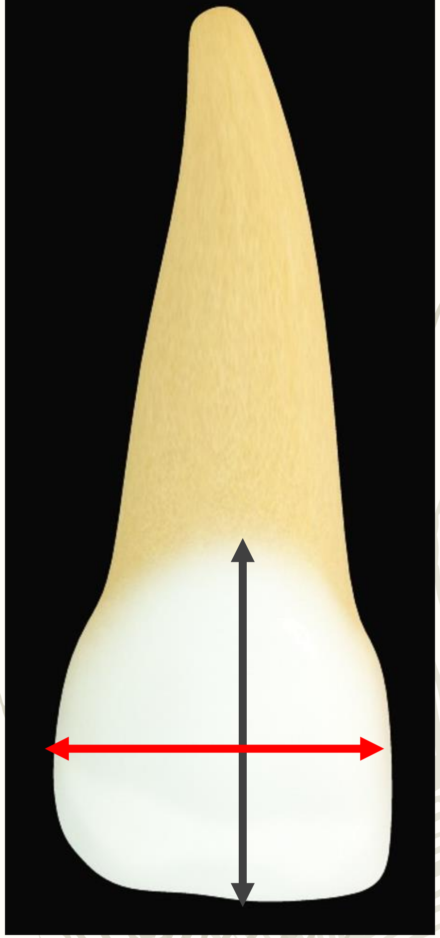
- 4 per arch
  - 2 centrals
  - 2 laterals
- Eruption
  - Mandibular before maxillary incisors
  - Central before lateral incisors
- **No mamelons, labial grooves or lobes.**
- The position of the incisors is usually relatively upright with spacing often between them.
- Attrition occurs, and a pattern of wear may be present.

DECIDUOUS MAXILLARY CENTRAL INCISORS

TOOTH	UNIVERSAL #	1ST EVIDENCE OF CALCIFICATION	CROWN COMPLETION	ERUPTION	ROOT COMPLETION
Deciduous Maxillary Central Incisor	E, F	14 weeks IU	1 ½ months	10 months	1 ½ years







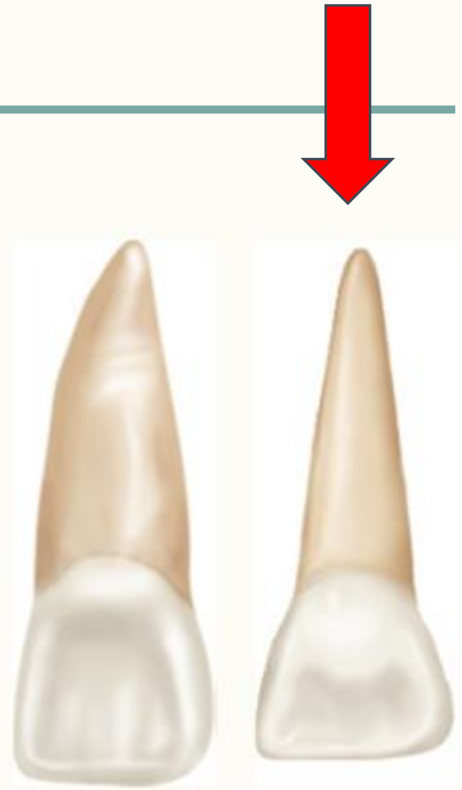
LABIAL ASPECT

DECIDIOUS MAXILLARY CENTRAL INCISORS

- **MD width greater than crown height** (what are the proportions b/w these two dimension in permanent counterparts)
- No labial grooves, depressions or lobes
- Smooth labial surface
- Straight incisal edge
- The root is cone-shaped with even, tapered sides
- Root length > Crown length as compared to Permanent CI.

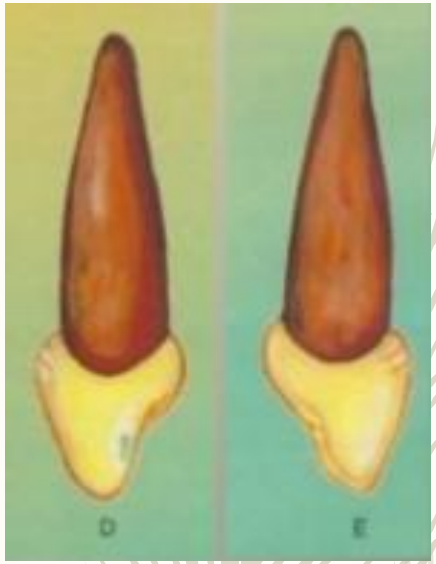
DECIDIOUS MAXILLARY CENTRAL INCISORS

- Well-developed marginal ridges
- **Well-developed developed cingulum which is;**
  - **Bulging & more incisally located**
  - **May extend further toward the incisal edge – lingual ridge and practically divide lingual surface into M & D fossae.**
  - **Unmarked by pits or grooves**
- Narrower root with a ridge on lingual surface as compared to labial (triangular in cross section)



LINGUAL ASPECT

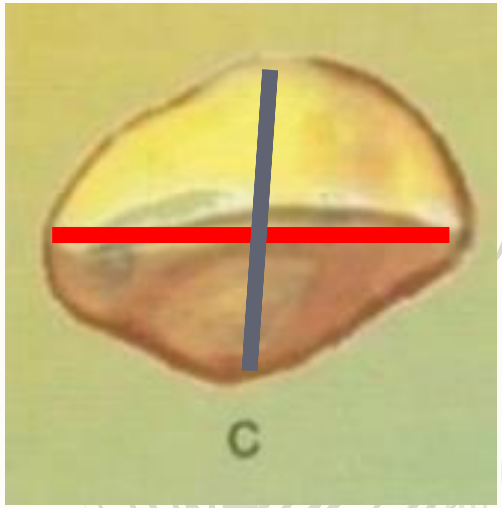
DECIDIOUS MAXILLARY CENTRAL INCISORS



MESIAL & DISTAL ASPECT

- Similar from both aspects
- Crown wider than total length as shown by measurement of crown at cervical 3rd.
- The curvature of the cervical line is curving distinctly towards the incisal ridge (Although curvature is less distinct than permanent counterpart).
- The cervical curvature distally is less than the curvature mesially.
- M & esp. D crown profiles overhang root profiles
- The root is cone-shaped with even, tapered sides, apex is blunt
- Usually Mesial has a developmental groove and distal is generally Convex

DECIDIOUS MAXILLARY CENTRAL INCISORS

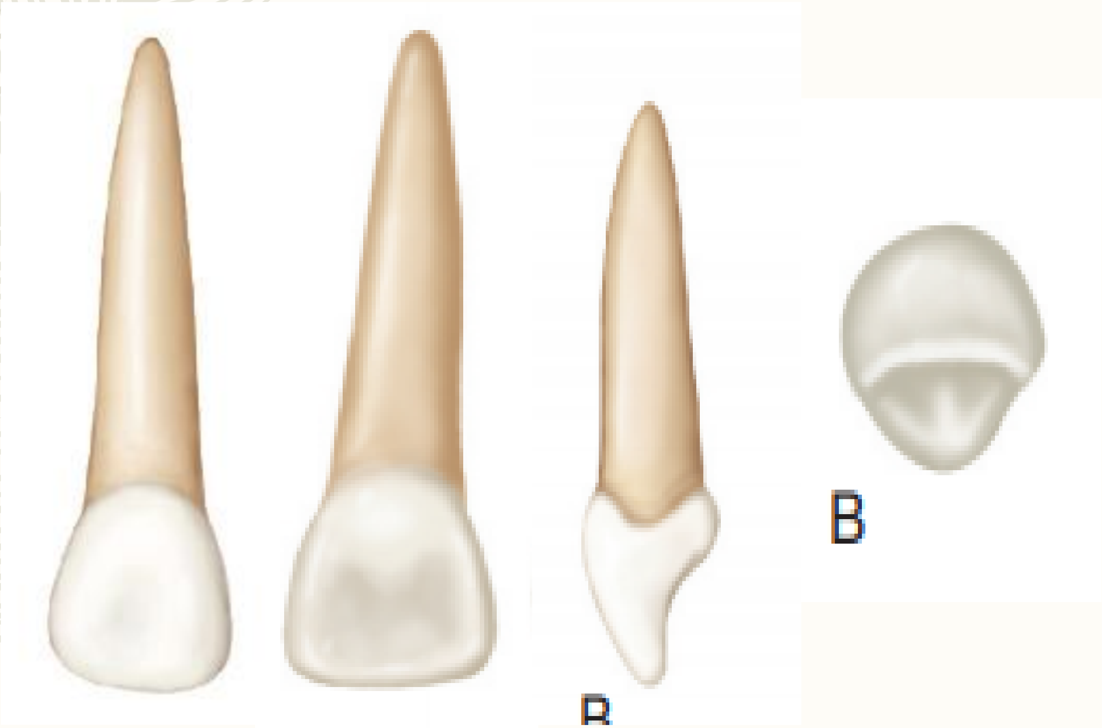


- **MD width > LL width prominent from this view.** (Recall for permanent?)
- **Incisal edge is centered over the main bulk of the crown and is relatively straight**
- The labial surface is much broader and also smoother than the lingual surface.
- The lingual surface tapers towards the cingulum.
- Generous dimensions mesially and distally at incisal 3rd prominent from this view lends good contacts with neighboring teeth at these points but only for a short time.

INCISAL ASPECT

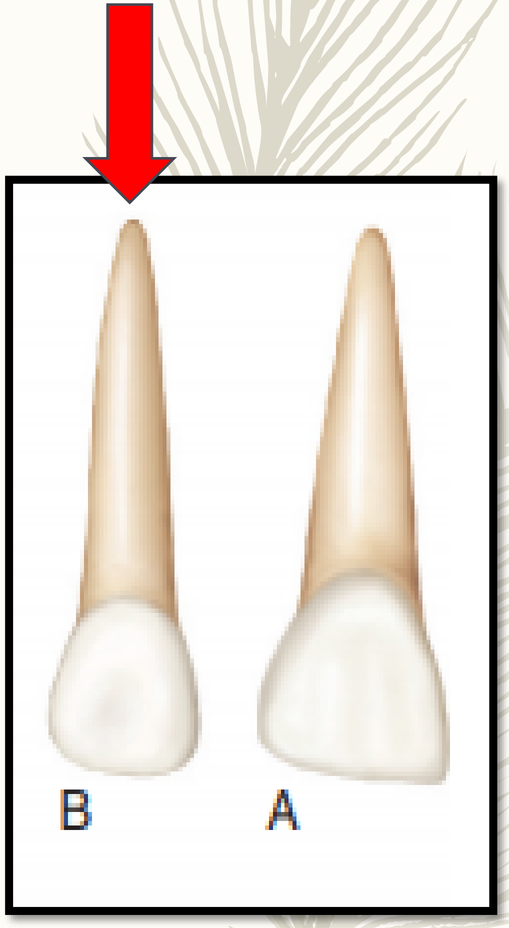
DECIDUOUS MAXILLARY LATERAL INCISORS

TOOTH	UNIVERSAL #	1ST EVIDENCE OF CALCIFICATION	CROWN COMPLETION	ERUPTION	ROOT COMPLETION
Deciduous Maxillary Lateral Incisor	D, G	16 weeks IU	2 ½ months	11 months	2 years



IFFERENCES BETWEEN PRIMARY MAXILLARY CENTRAL AND LATERAL INCISOR

– Similar to the Central Incisor from all aspects, but its dimensions differ.



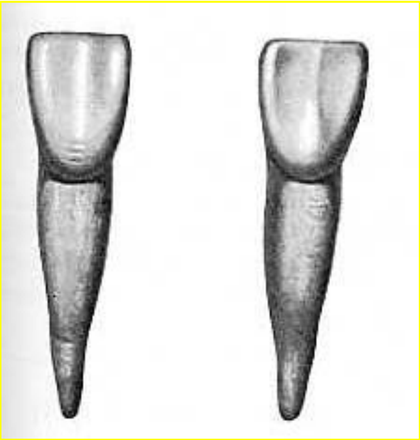
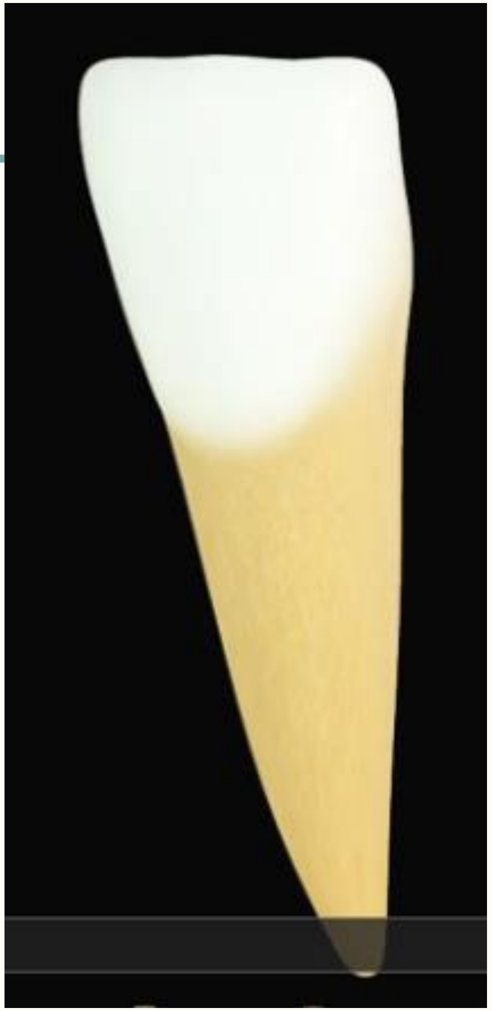
Differences Between Maxillary Central Incisor And Maxillary Lateral Incisor
Lateral Incisor crown < Central Incisor crown in all directions
Cervico-Incisal Width > Mesio Distal width of Maxillary Lateral Incisor
Disto-Incisal angle more rounded than those of Central Incisor
In contrast to Central Incisor, root of deciduous maxillary lateral incisor is much longer in proportion to its crown.

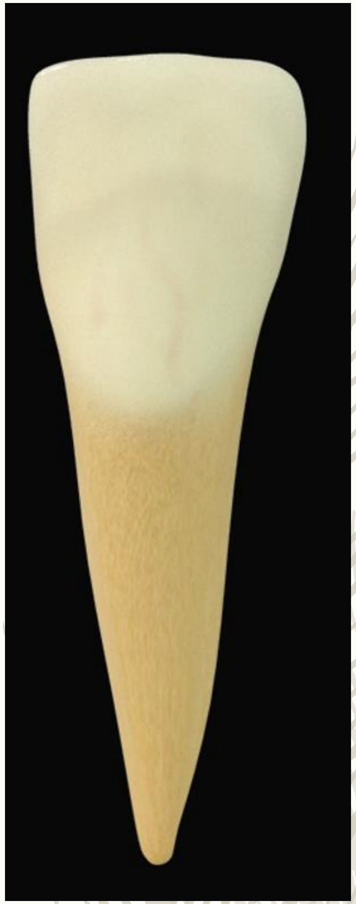
DECIDUOUS MANDIBULAR CENTRAL INCISOR

TOOTH	UNIVERSAL #	1ST EVIDENCE OF CALCIFICATION	CROWN COMPLETION	ERUPTION	ROOT COMPLETION
Deciduous Mandibular CI	P, O	14 weeks IU	2 ½ months	8 months	1 ½ years

- **Resembles Permanent Maxillary Lateral Incisor due to the heavy look at root trunk.**





DECIDUOUS MANDIBULAR CENTRAL INCISOR

- Flat surface with no developmental grooves
- Constricted cervix with M & D sides of crown tapered evenly from the contact areas.
- Cervical Line is similar to that of Max. incisors
- **90 degree MI & DI angles**
- Incisal edge is horizontal
- **Crown width > length as compared to permanent**
- Root
  - **2 times the height of the crown**
  - Narrow & conical with pointed apex

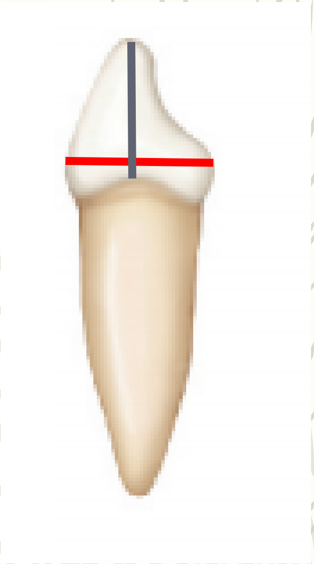
LABIAL ASPECT

DECIDUOUS MANDIBULAR CENTRAL INCISOR

- Prominent cingulum and marginal ridges
- A slight concavity or flattened surface which is at level with the marginal ridges called LINGUAL FIOSSAE is present on the Lingual surface at Middle and Incisal 3rd.
- **Lingual convergence of crown & root so lingual surface appears narrower than labial.**

LINGUAL ASPECT

DECIDUOUS MANDIBULAR CENTRAL INCISOR



MESIAL ASPECT

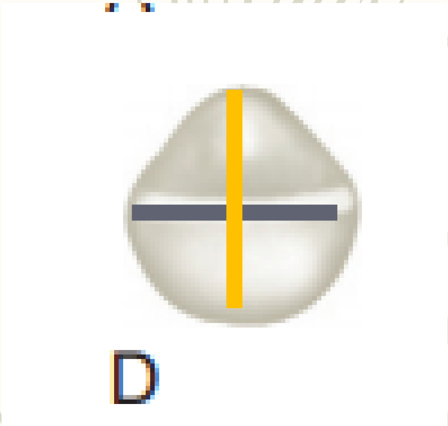
- M aspect shows typical incisor outline with smaller dimensions
- The incisal ridge is centered over the center of the root and between the crest of curvature of the crown, labially, and lingually.
- **Prominent cervical bulges Labially & Lingually in the Cervical 1/3rd, far more pronounced than permanent counterpart.**
- LL measurement is only about 1 mm < that of the primary maxillary CI.
- Root appears flat and evenly tapered; the apex presents a more blunt appearance than found with the lingual or labial aspects.

DECIDUOUS MANDIBULAR CENTRAL INCISOR

- Outline reverse of what could be seen from M aspect.
- Cervical Line of the crown is less curved toward the incisal ridge than on the mesial surface.
- Often, a developmental depression is evident on the distal side of the root.

DISTAL ASPECT

DECIDUOUS MANDIBULAR CENTRAL INCISOR

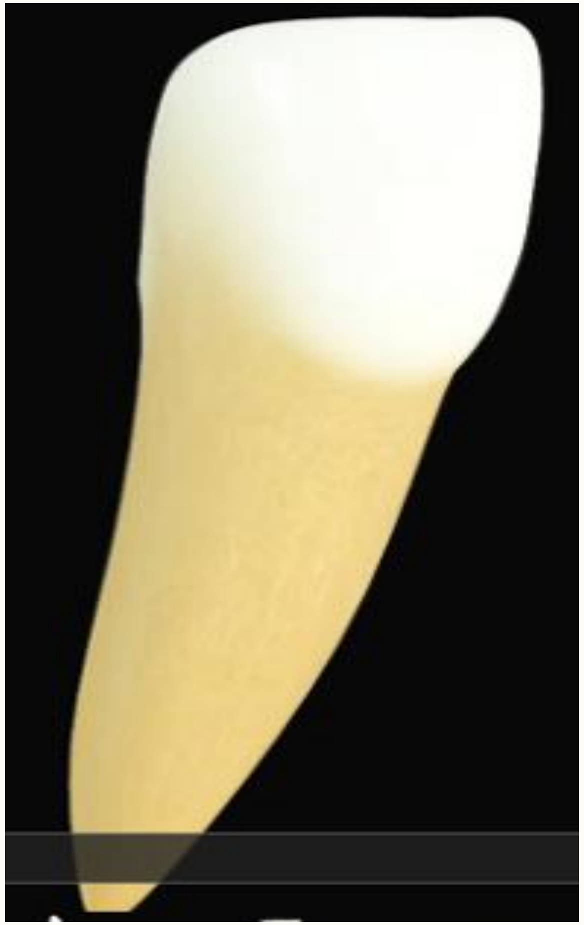


- Incisal ridge is straight and bisects the crown labiolingually
- Prominent crest of contours Labially and Lingually at their cervical 3rd.
- Evident Lingual convergence.
- The labial surface from this view presents a flat surface that is slightly convex, whereas the lingual surface presents a flattened surface that is slightly concave.

INCISAL ASPECT

DECIDUOUS MANDIBULAR LATERAL INCISOR

TOOTH	UNIVERSAL #	1ST EVIDENCE OF CALCIFICATION	CROWN COMPLETION	ERUPTION	ROOT COMPLETION
Deciduous Mandibular LI	Q, N	16 weeks IU	3 months	13 months	1 ½ years



SIMILARITIES & DIFFERENCES BETWEEN DECIDUOUS MANDIBULAR CENTRAL & LATERAL INCISOR

SIMILARITIES WITH MANDIBULAR CENTRAL INCISOR

- Supplements Primary Mandibular Central Incisor in function.
- Similar to Primary Mandibular Central Incisor
- Labio-Lingual Dimensions are equal

DIFFERENCES BETWEEN MANDIBULAR CENTRAL AND LATERAL INCISOR

- Lateral Incisors > Central Incisor in all dimensions.
- Cingulum more prominent than Mandibular Central Incisor.
- More concave Lingual Fossa.
- Incisal Ridge has tendency to slope downward distally.
- Distal Contact more Apical so good contact can be made with Primary Mandibular Canine.

Which of the following incisors present with lingual ridge?

A

B

C

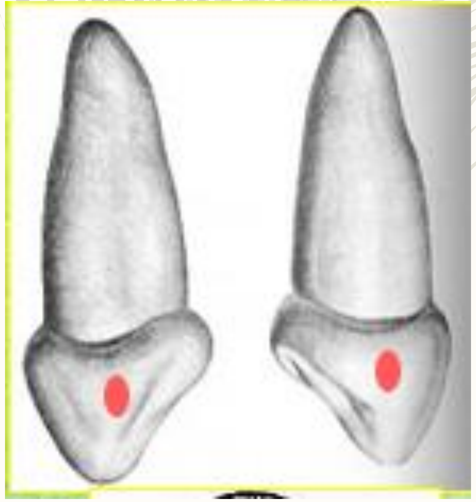
D

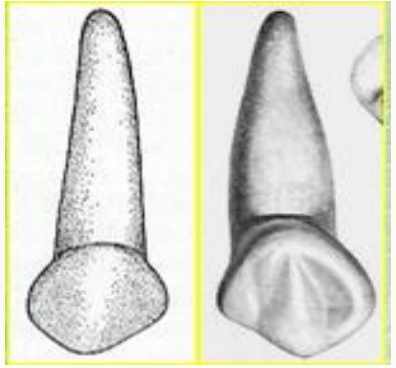
E

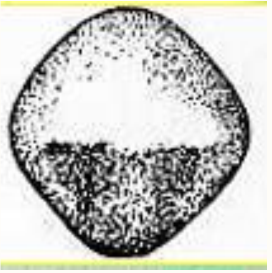
DECIDUOUS MAXILLARY CANINE

TOOTH	UNIVERSAL #	1ST EVIDENCE OF CALCIFICATION	CROWN COMPLETION	ERUPTION	ROOT COMPLETION
Deciduous Maxillary Canine	C, H	17 weeks IU	9 months	19 months	3 ¼ years

- Function of Deciduous Maxillary Canine : punch, tear, and apprehend food material
- Greater crown dimension along with root width and length, permits resistance against forces the tooth must withstand during function.

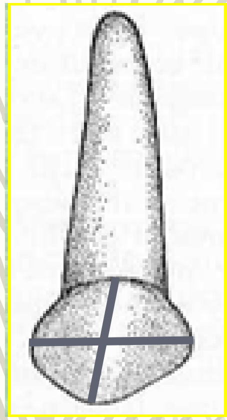






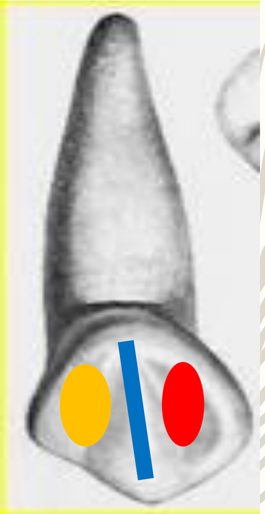
LABIAL ASPECT

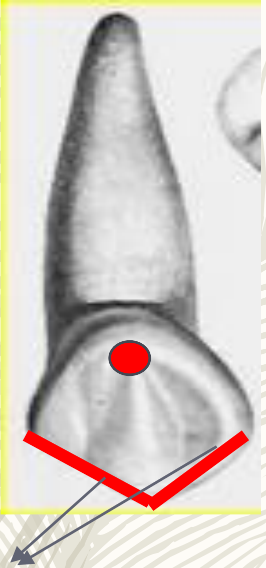
DECIDUOUS MAXILLARY CANINE



- **MD width > crown height.**
- constricted cervix in comparison to MD width
  - **Much more convex M & D surfaces as compared to either CI or LI.**
- **A long, well-developed, sharp cusp is present instead of a straight incisal edge.**
- **The cusp of Deciduous Maxillary canine is much longer and sharper than that of Permanent counterpart.**
- Mesial Crest of contour is more cervical than that of permanent canine.
- **M & D Contact areas at the same level such that a line drawn through the contact areas would bisect a line drawn from the cervix to the cusp tip, whereas in permanent canine Contact areas are not on the same level.**
- **Mesial slope of the cusp is longer than the distal slope**. (what about permanent counterpart?)
- The root is long, slender, and tapering and is more than twice the crown length.
- No labial ridge or depressions

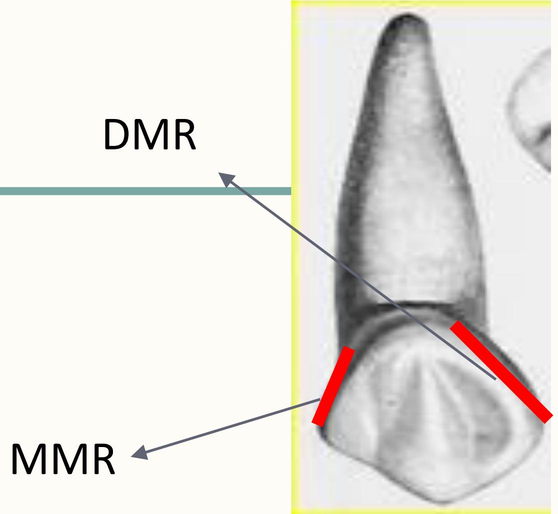
DECIDUOUS MAXILLARY CANINE





M & D CUSPAL RIDGES

LINGUAL ASPECT



- Pronounced enamel ridges merging with each other, including;
  - Cingulum
  - M & D marginal ridges
  - Incisal cuspal ridge
  - **Lingual ridge; divides the lingual surface into shallow ML & DL fossae**
  - **Tubercle at the cusp tip; continuation of the lingual ridge connecting the cingulum and the cusp tip**
- Root tapers lingually & is usually inclined distally also above the middle third

DECIDUOUS MAXILLARY CANINE

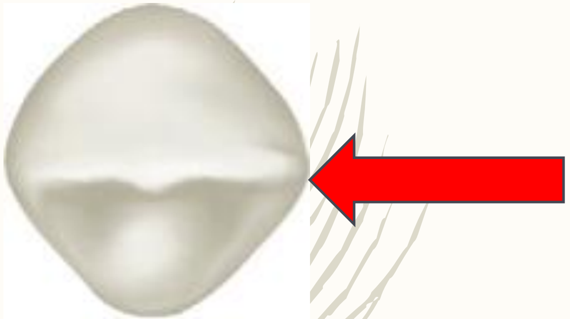
MESIAL & DISTAL ASPECT

MESIAL ASPECT

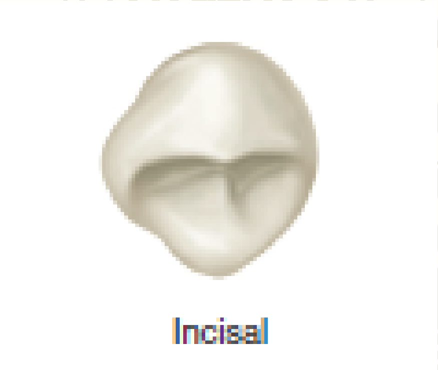
- Outline similar to CI & LI, different in proportions
- LL measurement in cervical 3rd much greater.

DISTAL ASPECT

- The distal outline is the reverse of the mesial aspect.
- Curvature of the cervical line toward the cusp ridge is less than on the mesial surface

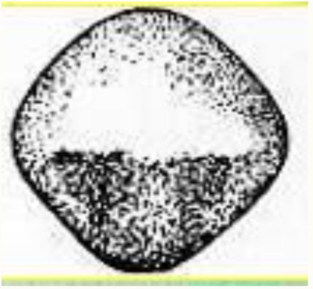


DECIDUOUS MAXILLARY CANINE



INCISAL ASPECT

- Essentially diamond-shaped
- **The following features are more pronounced and less rounded on Lingual aspects od Primary Maxillary Canines as compared to Permanent counterparts;**
  - **Angles formed at M & D contact areas**
  - **Cingulum on the lingual surface**
  - **The cervical third**
  - **Enamel ridge on the labial surface**
- **Cusp tip is distal to the center of the crown & M cusp slope is longer than the D cusp.**
- **Opposite for Mandibular Deciduous Canine, having D slope longer than M for inter-cuspation.**

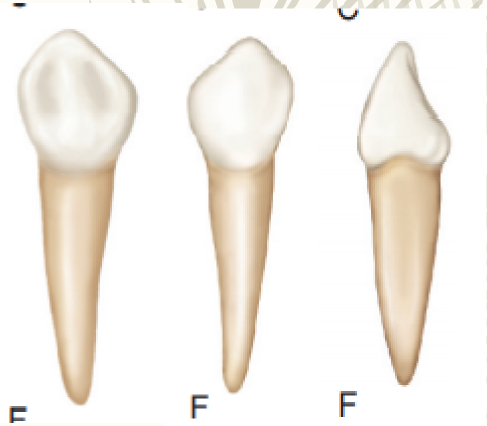


DECIDUOUS MANDIBULAR CANINE

TOOTH	UNIVERSAL #	1^{ST} EVIDENCE OF CALCIFICATION	CROWN COMPLETION	ERUPTION	ROOT COMPLETION
Deciduous Mandibular Canine	R, M	17 weeks IU	9 months	20 months	3 ¼ years

– Deciduous Mandibular Canine similar to Deciduous Maxillary Canine EXCEPT

Differences Between Maxillary & Mandibular Canine



- **CI width = 0.5 mm less than that of Maxillary Canine**
- **Root = 2 mm less than that of Maxillary Canine**
- **Mandibular Canine < Maxillary Canine in Labio-Lingual dimension**
- **MD measurement of root > MD measurement at contact points which contrasts with Maxillary Canine**
- **It is thicker at neck of tooth.**
- **The cervical ridges labially and lingually are not quite as pronounced as those found on the maxillary canine.**
- **Distal cuspal slope > Mesial cuspal slope (important for proper intercuspation)**

Table 3-1 Table of Measurements of the Primary Teeth of Man (Averages Only) (in Millimeters)

	Length Overall	Length of Crown	Length of Root	Mesiodistal Diameter of Crown	Mesiodistal Diameter of Crown at Cervix	Labiolingual Diameter of Crown	Labiolingual Diameter of Crown at Cervix
Upper Teeth							
Central incisor	16.0	6.0	10.0	6.5	4.5	5.0	4.0
Lateral incisor	15.8	5.6	11.4	5.1	3.7	4.0	3.7
Canine	19.0	6.5	13.5	7.0	5.1	7.0	5.5
Lower Teeth							
Central incisor	14.0	5.0	9.0	4.2	3.0	4.0	3.5
Lateral incisor	15.0	5.2	10.0	4.1	3.0	4.0	3.5
Canine	17.5	6.0	11.5	5.0	3.7	4.8	4.0

Which of the following canines presents with a longer mesial slope?

A

B

C

D

THANK

YOU!